



Student Number : _____

YEAR 12

TRIAL HIGHER SCHOOL CERTIFICATE

EXAMINATION 2021

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper (pages 21-24)
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I – 10 marks (pages 2 – 8)

- Attempt Questions 1 – 10
- Answer on the multiple choice sheet provided on page 19 (you may detach this sheet)
- Allow about 15 minutes for this section

Section II – 60 marks (pages 9 – 18)

- Attempt Questions 11 – 14
- Answer each question in a separate answer booklet
- Allow about 1 hours and 45 minutes for this section

Section 1 – 10 Marks (1 mark each)

- Attempt Questions 1-10
 - Answer on the Multiple Choice Sheet provided on Page 19
(You may detach this sheet)
 - Allow about 15 minutes for this section
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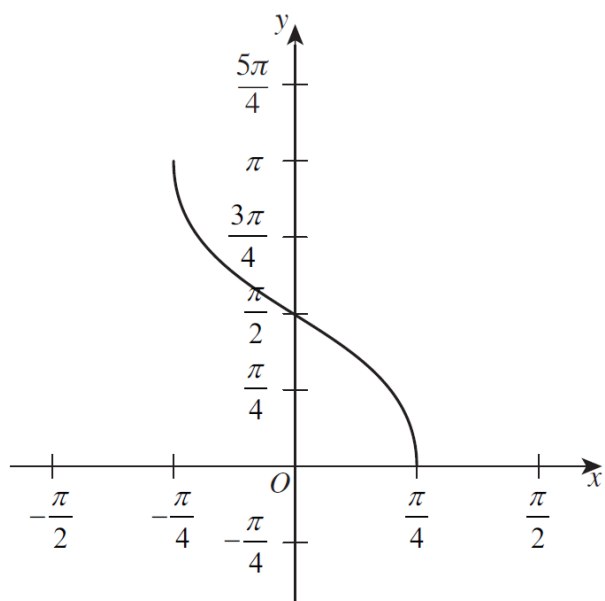
1. Which of the following is NOT a factor of the polynomial
 $P(x) = x^3 - x^2 - 8x + 12$?

- (A) $(x + 3)$
- (B) $(x + 3)^2$
- (C) $(x - 2)$
- (D) $(x - 2)^2$
-

2. What is the primitive of $\frac{\ln 2}{\sqrt{\pi - x^2}}$?

- (A) $(\ln 2)(x) \left(\sin^{-1} \left(\frac{x}{\sqrt{\pi}} \right) \right) + C$
- (B) $(\ln 2) \left(\sin^{-1} \left(\frac{x}{\sqrt{\pi}} \right) \right) + C$
- (C) $\left(\frac{\ln 2}{\sqrt{\pi}} \right) \left(\sin^{-1} \left(\frac{x}{\sqrt{\pi}} \right) \right) + C$
- (D) $(\ln 2) \left(\sin^{-1} \left(\frac{x}{\pi} \right) \right) + C$

3. The graph of an inverse trigonometric function is shown.



What is the equation of this function?

(A) $\frac{4}{\pi} \cos^{-1} x$

(B) $\frac{\pi}{4} \cos^{-1} x$

(C) $\cos^{-1} \left(\frac{4x}{\pi} \right)$

(D) $\cos^{-1} \left(\frac{\pi x}{4} \right)$

4. A bag contains 2 red, 5 blue, 6 white, 11 green and 14 yellow marbles. What is the minimum number of marbles that need to be chosen randomly from the bag to ensure that 6 marbles of the same colour have been chosen?

(A) 16

(B) 17

(C) 23

(D) 34

-
5. The function $f(x) = x^2 - 4x + 7$ has an inverse function $f^{-1}(x)$ in the domain $x \geq 2$.

What is the value of $f^{-1}(f(m))$,
where m is a real number NOT in the domain $x \geq 2$?

(A) m

(B) $2 - m$

(C) $2 + m$

(D) $4 - m$

6. The vectors $4\hat{i} - \hat{j}$ and $3\hat{i} + m\hat{j}$ are perpendicular.
What is the value of m ?

- (A) -12
- (B) -1
- (C) 3
- (D) 12

7. A piece of hot metal is placed in a room with a surrounding air temperature of 15°C and is allowed to cool.
It loses heat according to Newton's Law of Cooling:

$$\frac{dT}{dt} = -k(T - A) \text{ where}$$

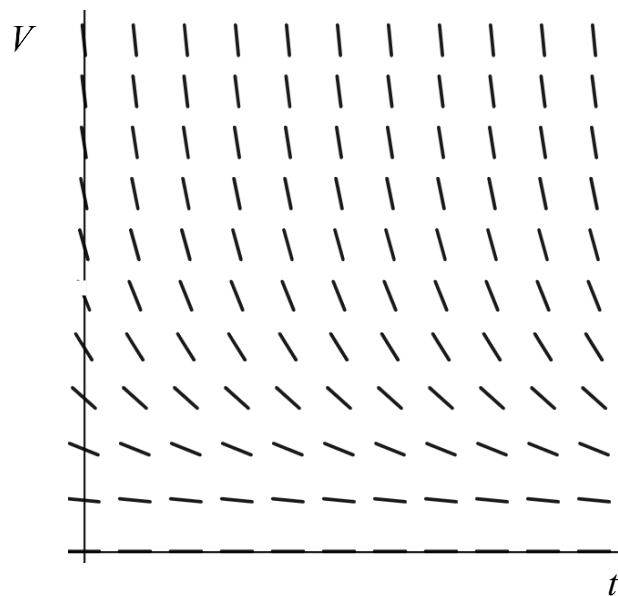
- T is the temperature of the metal in $^{\circ}\text{C}$, at time t minutes
- A is the surrounding air temperature
- k is a positive constant

After 5 minutes the temperature of the metal is 75°C
and after a FURTHER 3 minutes the temperature of the metal is 45°C .

What is the value of k in the above equation?

- (A) $3\ln(0.5)$
- (B) $3\ln(2)$
- (C) $\frac{\ln(0.5)}{3}$
- (D) $\frac{\ln(2)}{3}$

8. A direction field for the volume of water, V megalitres, in a reservoir t years after 2021 is shown below.



According to this model, for $k > 0$, what is the value of $\frac{dV}{dt}$?

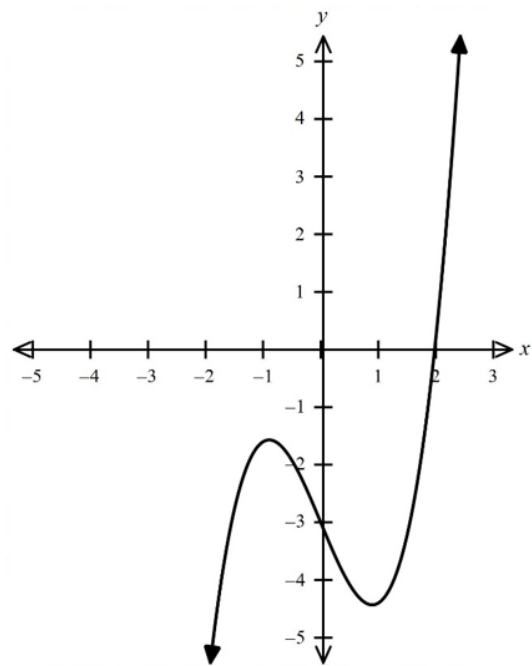
(A) $-kV^2$

(B) $-\frac{k}{V}$

(C) $\frac{k}{V}$

(D) kV^2

9. Consider the graph of $y = f(x)$ shown below:



Which of the following would have two MORE roots than $f(x)$?

(A) $y = -2 \times f(x)$

(B) $y = f(x) + 3$

(C) $y = f^{-1}(x)$

(D) $y = f(x + 3)$

10. It is known that $\sin x = \frac{1}{3}$, where $\frac{\pi}{2} < x < \pi$

What is the value of $\cos(2x)$?

(A) $-\frac{7}{9}$

(B) $\frac{7}{9}$

(C) $\frac{2}{3}$

(D) $-\frac{4\sqrt{2}}{3}$

END OF SECTION 1

SECTION 2 – 60 Marks

- Attempt Questions 11-14
- Allow about 1h 45min for this section
- Answer each question in a separate writing booklet (extra writing booklets are available upon request from the examination supervisor)
- Show all relevant mathematical reasoning and/or calculations

Marks

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) (i) Write $\cos(7x)\sin(5x)$ as a sum/difference of trigonometric ratios 1

(ii) Hence, solve $2\cos(7x)\sin(5x) = \sin(12x) - \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$ 3

(b) Prove by mathematical induction that $2^{2n} + 6n - 1$ is divisible by 3 for all integers $n \geq 1$ 3

(c) Consider the polynomial $P(x) = x^3 - 2px + q$ where p and q are constants and $p \neq 0$.

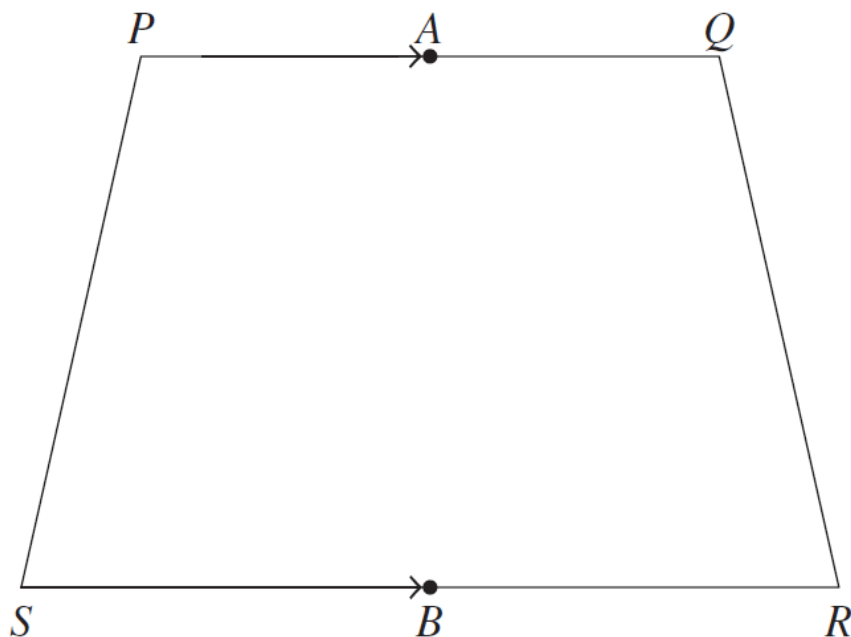
It is given that α , β and $\alpha + \beta$ are roots of the equation $P(x) = 0$.

(i) Prove that $\alpha = -\beta$ 1

(ii) Hence, or otherwise, find all the roots of $P(x) = 0$ in terms of p . 3

Question 11 continues on page 10

- (d) $PQRS$ is a trapezium with A and B being the midpoints of the parallel sides PQ and RS respectively. Marks



Let $\overrightarrow{PA} = \underline{a}$ and $\overrightarrow{SB} = \underline{b}$.

- (i) Express $\overrightarrow{QR}$ in terms of $\underline{a}$, $\underline{b}$ and $\overrightarrow{AB}$

2

- (ii) Hence, or otherwise, show that $\overrightarrow{AB} = \frac{1}{2}(\overrightarrow{PS} + \overrightarrow{QR})$

2

END OF QUESTION 11

Question 12 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) (i) Write the expression $\sqrt{3}\sin x + \cos x$ in the form $R\sin(x + \alpha)$,
where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

2

- (ii) Write down the three missing numbers that would appear in the cells of the table below, where $g(x) = \sqrt{3}\sin x + \cos x = R\sin(x + \alpha)$ and the numbers are given correct to 2 decimal places

1

x	$-\frac{\pi}{6}$	0	$\frac{2\pi}{6}$	$\frac{5\pi}{6}$	$\frac{8\pi}{6}$	$\frac{11\pi}{6}$	$\frac{14\pi}{6}$	$\frac{17\pi}{6}$	$\frac{20\pi}{6}$
$\ln x$			0.05	0.96	1.43	1.75			2.35
$g(x)$	0	1	2	0	-2	0	2	0	-2

- (iii) Sketch the graphs of $y = \ln x$ and $y = g(x)$ on the same number plane, showing clearly any point(s) of intersection.
(You do not need to give the coordinates of any point(s) of intersection, just show their location on your sketch)

3

- (iv) Hence, determine the number of solutions to the equation $\sqrt{3}\sin x + \cos x = \ln x$ in the domain $(0, \infty)$

1

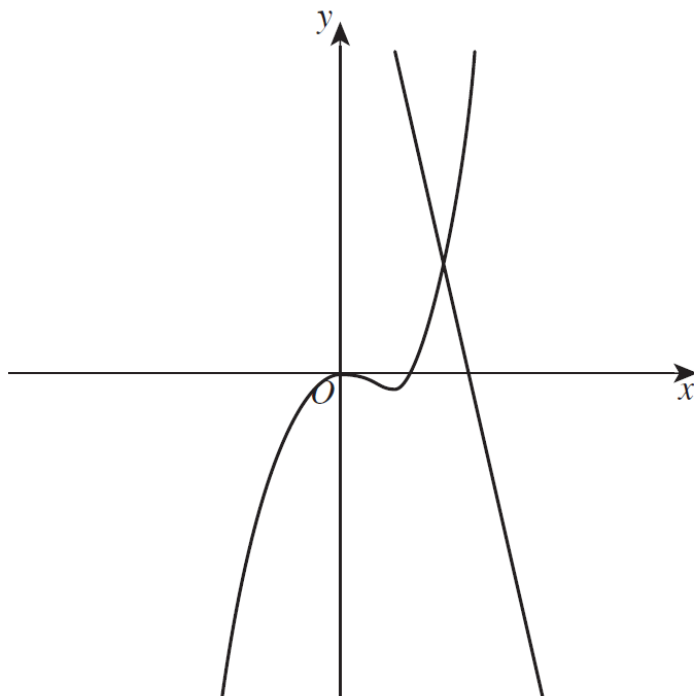
- (b) Use the identity $(1+x)^m(1+x)^n = (1+x)^{m+n}$ to show that

3

$$\binom{m+n}{4} = \binom{n}{4} + \binom{n}{3}\binom{m}{1} + \binom{n}{2}\binom{m}{2} + \binom{n}{1}\binom{m}{3} + \binom{m}{4}$$

Question 12 continues on page 12

- (c) The graphs of $g(x) = mx + b$ and $f(x) = 3x^3 - 2x^2$ are shown below



The solution to the equation $mx + b < 3|x|^3 - 2|x|^2$ is $x \in (-\infty, -2) \cup (1, \infty)$.

- (i) Sketch the graph of $y = 3|x|^3 - 2|x|^2$ 1
- (ii) Hence, find the values of m and b . 3
- (d) In how many different ways can 6 runners finish 1st, 2nd and 3rd in a race? 1
(assume no equal placings)

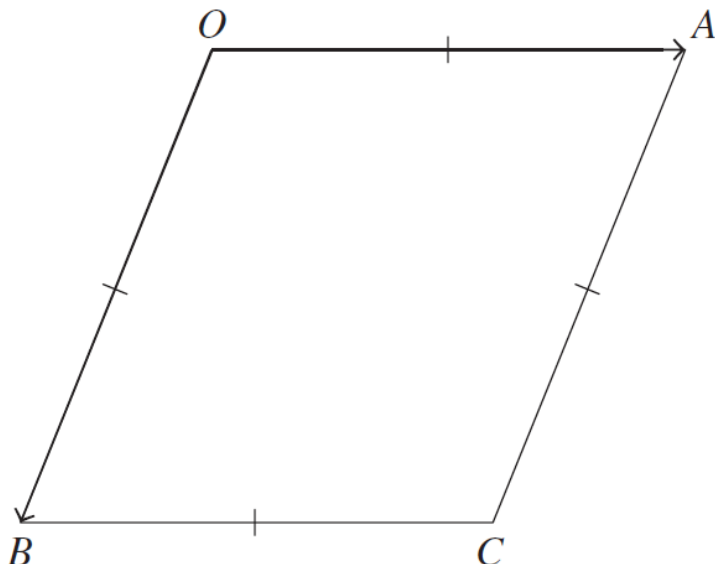
END OF QUESTION 12

Question 13 (15 marks) Use a SEPARATE writing booklet

Marks

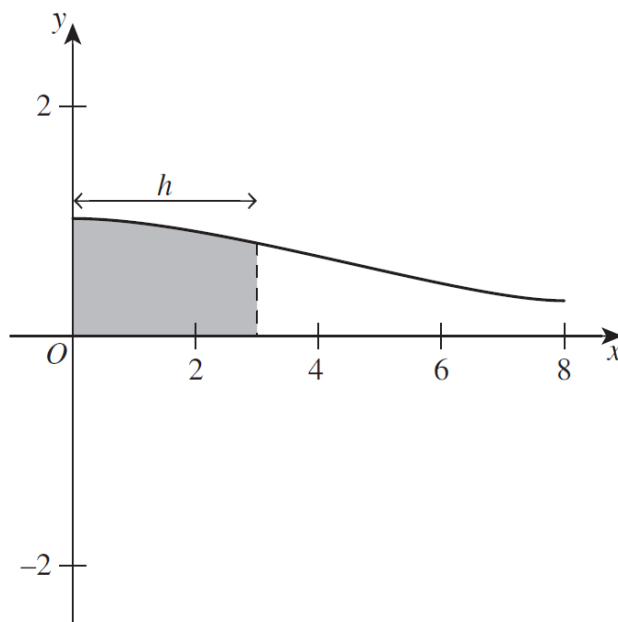
- (a) $OACB$ is a rhombus with $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$

2



Using vector methods, prove that the diagonal OC bisects angle AOB .

- (b) The graph of $y = \frac{3}{\sqrt{9+x^2}}$ for $x > 0$ is shown below.

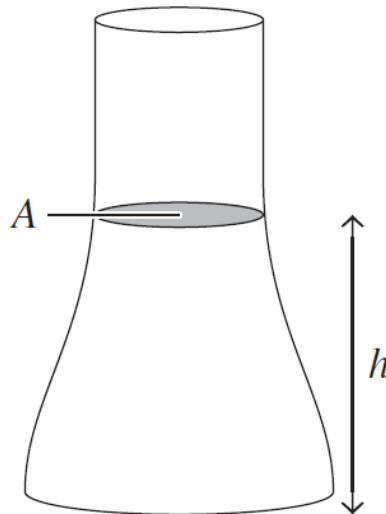


Question 13(b) continues on page 14

- (i) The area bounded by $y = \frac{3}{\sqrt{9+x^2}}$,
the x -axis and the lines $x=0$ and $x=h$ is rotated about the x -axis
to create a solid of revolution. 2

Show that the volume of the solid is given by $V = 3\pi \tan^{-1}\left(\frac{h}{3}\right)$ cubic units.

- (ii) The solid of revolution of the graph has the shape of a vase lying down
on its side. 2
The diagram shows a vase of the same shape standing upright.



As water is being poured into the vase, the height h of the water
in the vase is increasing at a rate of 3 cm/s .

Find the exact rate at which the volume of water in the vase,
 V , is increasing when $h = 6$.

- (iii) The surface of the water forms a circular shape with area A .
By using the original graph, or otherwise, find the exact rate at which the
area, A , is decreasing when $h = 6$. 2

Question 13 continues on page 15

- (c) The spread of flu through a student population is modelled by the equation

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$
 where S is the total number of students infected after t days

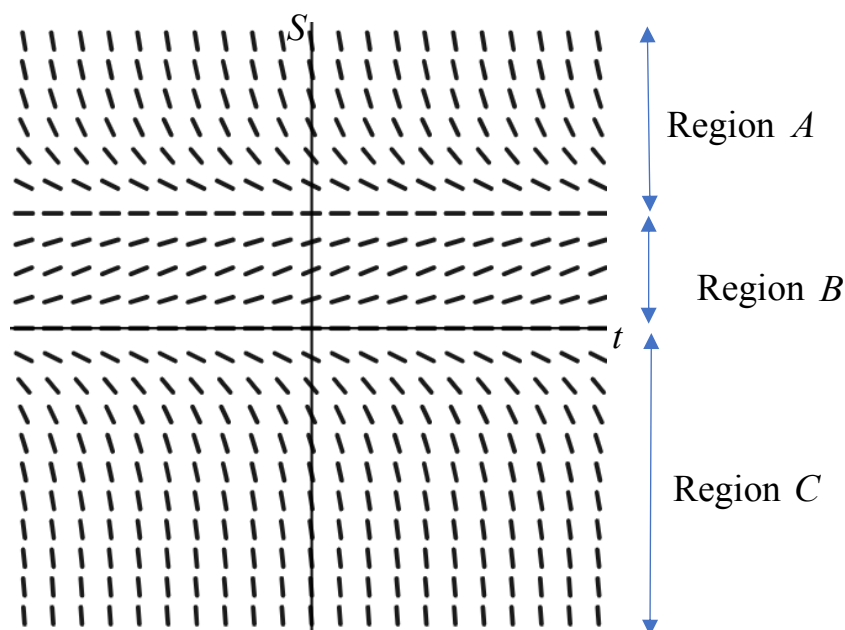
- (i) Show that the given equation for S satisfies the differential equation

$$\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$$

3

- (ii) The slope field of $\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$ is shown.

2



There are 3 regions labelled A , B and C in which a solution curve can be found.

In which one of these three regions can the solution curve exist?
 Justify your answers with reference to constant solutions and the initial value of S .

- (iii) According to this model, after how many days does the rate of increase reach a maximum? Give your answer to the nearest number of days.

2

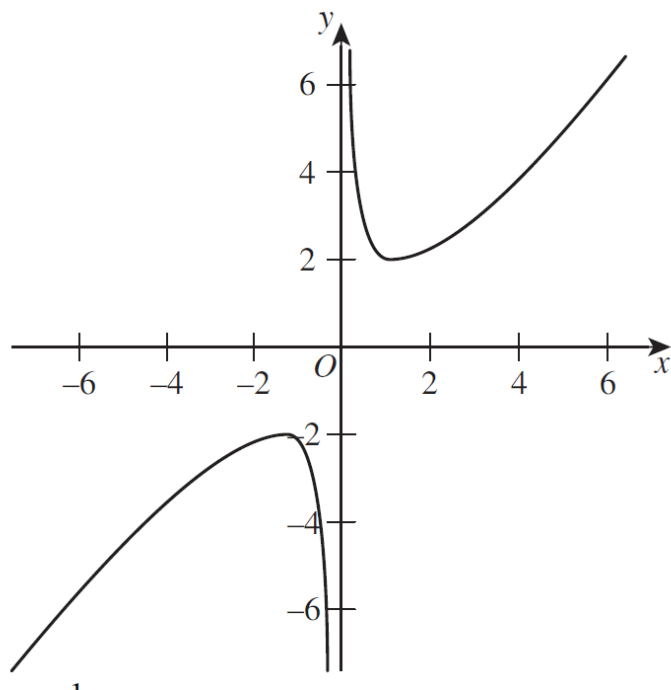
END OF QUESTION 13

Question 14 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) The graph of the function $f(x) = \frac{x^2 + 1}{x}$ is shown

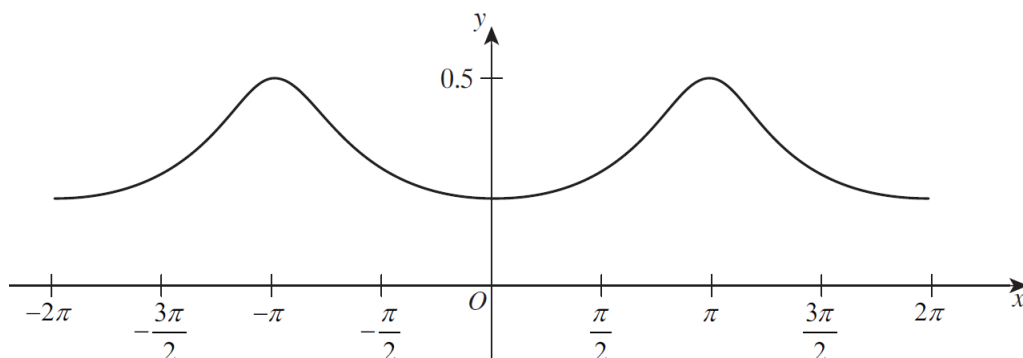
3



Sketch the graph of $y = \frac{1}{\sqrt{f(x)}}$ showing all important features including turning point(s), intercept(s) and asymptote(s).

Question 14 continues on page 17

- (b) The graph of the function $y = \frac{1}{5 + 3\cos x}$ for $-2\pi \leq x \leq 2\pi$ is shown below:



- (i) By using the substitution $t = \tan\left(\frac{x}{2}\right)$ and t – formula(s), show that 3

$$\int \frac{1}{5 + 3\cos x} dx = \frac{1}{2} \tan^{-1}\left(\frac{1}{2} \tan\left(\frac{x}{2}\right)\right) + C \text{ where } C \text{ is a constant.}$$

- (ii) Hence, show that the area bounded by the curve $y = \frac{1}{5 + 3\cos x}$, 2
the lines $x = 0$, $x = \pi$ and the x – axis is $\frac{\pi}{4}$

You may use the substitution $\tan \frac{\pi}{2} = \infty$ if necessary.

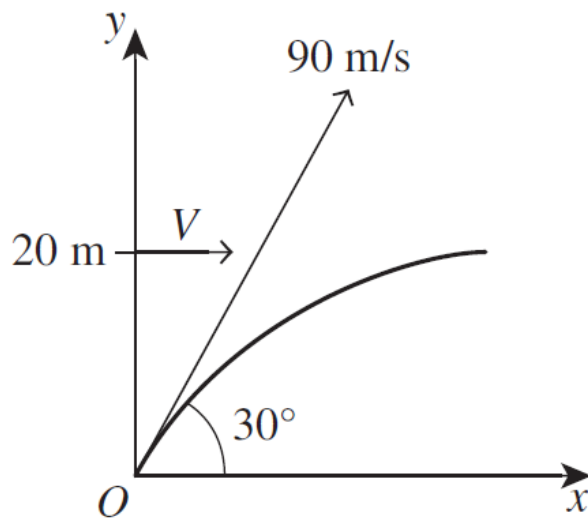
Question 14 continues on page 18

- (c) (i) Initially a golf ball was hit from the ground at 90 ms^{-1} at an angle of 30° to the horizontal and $g = 10 \text{ ms}^{-2}$. 2

Show that the position vector of the golf ball after t seconds is given by

$$\underline{s}(t) = (45\sqrt{3}t)\underline{i} + (45t - 5t^2)\underline{j}$$

- (ii) Five seconds after the golf ball was hit, a small stone was fired with velocity V horizontally from a point 20 m above the ground. 3



By finding the position vector of the stone, or otherwise, show that the two objects collide after the golf ball has travelled for 21 seconds.

- (iii) What is the stone's speed at collision?
Give your answer to 3 significant figures. 2

END OF EXAMINATION

$$1/ \quad P(2) = (2)^3 - (2)^2 - 8(2) + 12 = 0$$

$\therefore (x-2)$ is a factor

$$P(-3) = (-3)^3 - (-3)^2 - 8(-3) + 12 = 0$$

$\therefore (x+3)$ is a factor

Product of roots is -12

$$2^2 \times (-3) = -12$$

$\therefore (x-2)^2$ is a factor

so $(x+3)^2$ is not a factor

[B]

Note: Multiple choice logic

If $x+3$ was not a factor then $(x+3)^2$ wouldn't be.

Similarly for $x-2$ + $(x-2)^2$.

This means $(x+3)$ + $(x-2)$ are factors ... we can only eliminate one of the four options so it must be $(x-2)^2$ or $(x+3)^2$.

$$2^2 \times (-3) = -12$$

$$2 \times (-3)^2 = 18$$

$\therefore (x+3)^2$ is NOT

$\therefore$ **[B]**

①

$$2/ \int \frac{\ln 2}{\sqrt{\pi - x^2}} dx$$

$$= \ln 2 \int \frac{1}{\sqrt{(\sqrt{\pi})^2 - (x)^2}} dx$$

Using $\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1}\left(\frac{f(x)}{a}\right) + C$ we get

$$I = \ln 2 \cdot \sin^{-1} \frac{x}{\sqrt{\pi}} + C$$

$\therefore$ B

or differentiate each option

$$\text{eg A) } \ln 2 \times \left[x \cdot \frac{\frac{1}{\pi}}{\sqrt{1 - \left(\frac{x}{\pi}\right)^2}} + \sin^{-1} \frac{x}{\pi} \cdot 1 \right]$$

$$= \ln 2 \cdot x \cdot \frac{\frac{1}{\pi}}{\frac{\sqrt{\pi^2 - x^2}}{\pi}} + \sin^{-1} \frac{x}{\pi}$$

$$= \frac{\ln 2}{\sqrt{\pi^2 - x^2}} + \sin^{-1} \frac{x}{\pi} \quad \times$$

etc

3. Normally for $y = \cos^{-1} x$ $-1 \leq x \leq 1$, $0 \leq y \leq \pi$

The range hasn't changed but now $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

so

$$-1 \leq \cos x \leq 1$$

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

$$-\pi \leq 4x \leq \pi$$

$$-1 \leq \frac{4}{\pi} x \leq 1$$

$$\therefore y = \cos^{-1}\left(\frac{4x}{\pi}\right)$$



4. You could choose

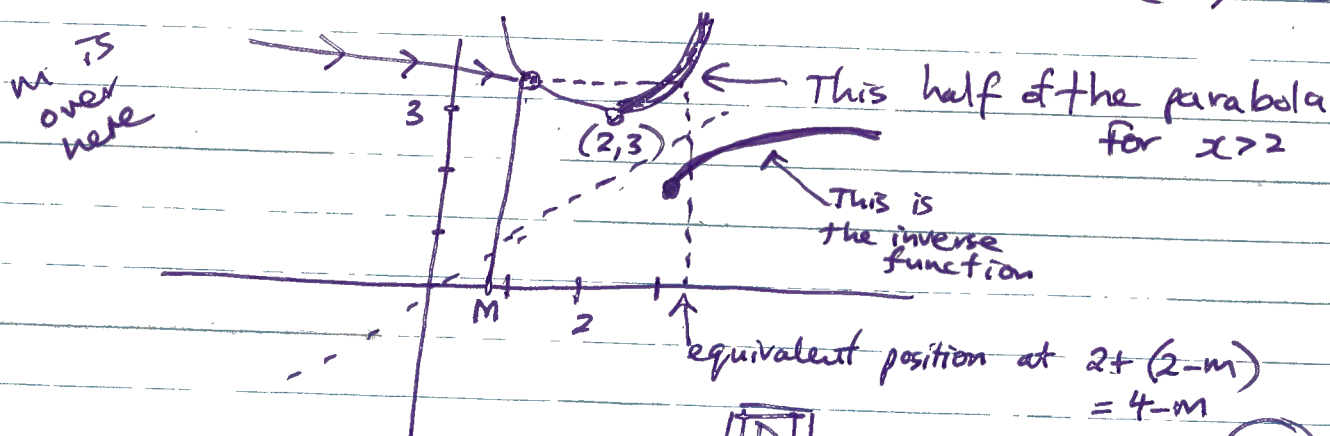
2 Red + 5 blue + 5 white + 5 green + 5 yellow
and then the next marble will make
6 of one colour.

$$\therefore 23$$



5.

$$x^2 - 4x + 7 = x^2 - 4x + 2^2 - 2^2 + 7 = (x-2)^2 + 3$$



$$6/ \quad (4)(3) + (-1)(m) = 0$$

$$m = 12$$

(dot product is zero for $\perp$ vectors)



$$7/ \quad \frac{dT}{dt} = -k(T-15)$$

from 1st sentence + second dot point

$$\frac{dT}{T-15} = -k dt$$

$$\ln(T-15) = -kt + c$$

$$\log_e(T-15) = -kt + c$$

(note: $\ln|T-15| = \ln(T-15)$ since $T > 15$)

$$e^{-kt+c} = T-15$$

$$me^{-kt} = T-15$$

(let $m = e^c$)

$$T = me^{-kt} + 15$$

when $t = 5$ $T = 75$

when $t = 8$ $T = 45$

$$75 = me^{-5k} + 15$$

$$45 = me^{-8k} + 15$$

$$60 = me^{-5k} \quad \text{--- (1)}$$

$$30 = me^{-8k} \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2}$$

$$2 = \frac{e^{-5k}}{e^{-8k}}$$

$$2 = e^{3k}$$

$$\ln 2 = \ln e^{3k}$$

$$3k = \ln 2$$

$$k = \frac{\ln 2}{3}$$



8/ $\frac{dv}{dt}$ is the gradient

From the direction field the gradient is always negative eg when $V=10$

Eliminate $\rightarrow C + D$ are positive

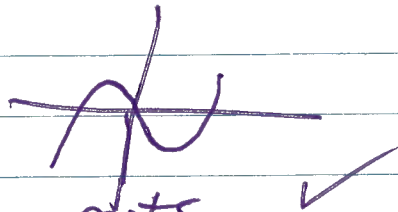
Now as V increases $\frac{dv}{dt}$ is more negative (greater in size)

The opposite would happen for B

$\therefore$ A

9/ A) Reflect in x -axis + make all points twice as far from x -axis
 $\rightarrow$ still only 1 root

B) Translate curve up 3 units then



Yes 2 more roots ✓

C) Reflect in line $y=x$

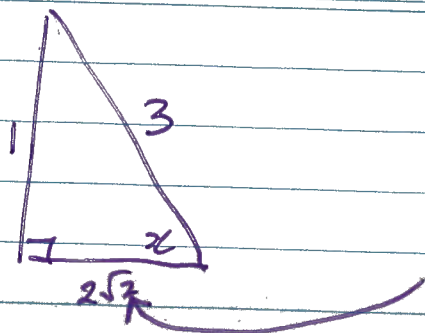


coords "swap" $\therefore$ only root is at $(-3, 0)$

D) Translate 3 units left .. still 1 root

$\therefore$ B

10/



By Pythagoras

$$\sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}$$

Now x is obtuse, so $\cos x = -\frac{2\sqrt{2}}{3}$

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= \left(-\frac{2\sqrt{2}}{3}\right)^2 - \left(\frac{1}{3}\right)^2\end{aligned}$$

$$= \frac{8 - 1}{9}$$

$$= \frac{7}{9}$$

B

$$\text{1/ a) i) } \cos 7x \sin 5x = \frac{1}{2} [\sin(7x+5x) - \sin(7x-5x)] \\ = \frac{1}{2} [\sin 12x - \sin 2x]$$

$$\text{ii) } 2 \cos 7x \sin 5x = \sin 12x - \frac{\sqrt{3}}{2}$$

$$2 \times \frac{1}{2} [\sin 12x - \sin 2x] = \sin 12x - \frac{\sqrt{3}}{2}$$

$$\sin 12x - \sin 2x = \sin 12x - \frac{\sqrt{3}}{2}$$

$$\sin 2x = \frac{\sqrt{3}}{2}$$

$$2x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$$

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}$$

b) Show true for $n=1$

$$2^{2(1)} + 6(1) - 1 = 4 + 6 - 1 \\ = 9 \\ = 3 \times 3$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\overset{ie}{2^{2k}} + 6k - 1 = 3M, \text{ M is an integer}$$

$$\text{rearranging } 2^k = 3M - 6k + 1 \quad *$$

Prove true for $n=k+1$ if true for $n=k$

ie RTP $2^{k+1} + 6(k+1) - 1 = 3Q$ Q is an integer

$$\begin{aligned} \text{LHS} &= 2^{2k+2} + 6k+6 - 1 \\ &= 2^{2k} \times 2^2 + 6k + 5 \\ &= (3M - 6k + 1) \times 4 + 6k + 5 \text{ from } * \end{aligned}$$

$$\begin{aligned} &= 12M - 24k + 4 + 6k + 5 \\ &= 12M - 18k + 9 \\ &= 3(4M - 6k + 3) \\ &= 3Q \text{ where } Q = 4M - 6k + 3 \\ &\text{and } Q \text{ is an integer since } 4, M, 6, k, 3 \text{ are integers} \end{aligned}$$

$\therefore 2^{k+1} + 6(k+1) - 1$ is divisible by 3
if $2^k + 6k - 1$ is divisible by 3

Conclusion

Since $2^1 + 6(1) - 1$ is divisible by 3, and $2^{k+1} + 6(k+1) - 1$ is divisible by 3 if $2^k + 6k - 1$ is divisible by 3, by the principle of mathematical induction the statement is true for all integers ≥ 1 .

c) $P(x) = x^3 + 0x^2 - 2px + q$

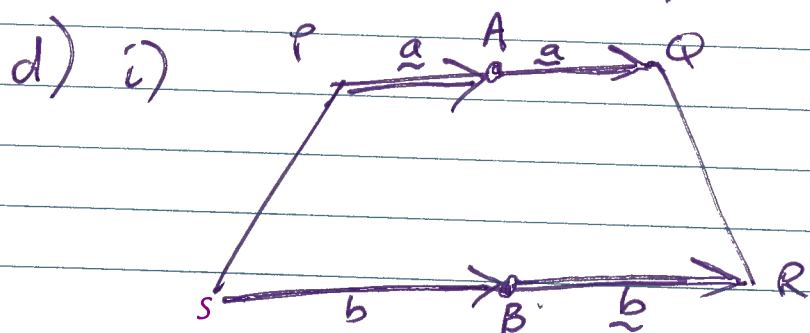
i) sum of roots is 0

$$\begin{aligned} \therefore \alpha + \beta + (\alpha + \beta) &= 0 \\ 2\alpha + 2\beta &= 0 \\ 2\alpha &= -2\beta \\ \alpha &= -\beta \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \alpha\beta + \alpha(\alpha+\beta) + \beta(\alpha+\beta) &= -2p \\
 \alpha(-\alpha) + \alpha^2 + \alpha(-\alpha) + (-\alpha)(\alpha) + (-\alpha)(-\alpha) &= -2p \\
 -\alpha^2 + \alpha^2 - \alpha^2 - \alpha^2 + \alpha^2 &= -2p \\
 -\alpha^2 &= -2p \\
 \alpha^2 &= 2p \\
 \alpha &= \pm\sqrt{2p}
 \end{aligned}$$

$$\therefore \alpha + \beta = \sqrt{2p} - \sqrt{2p} = 0$$

The roots are $-\sqrt{2p}$, $\sqrt{2p}$ and 0.



$$\begin{aligned}
 \vec{QR} &= \vec{QA} + \vec{AB} + \vec{BR} \\
 &= -\vec{a} + \vec{AB} + \vec{b} \\
 &= \vec{AB} + \vec{b} - \vec{a}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \vec{PS} &= \vec{PA} + \vec{AB} + \vec{BS} \\
 &= \vec{a} + \vec{AB} - \vec{b} \\
 &= \vec{AB} + (\vec{a} - \vec{b})
 \end{aligned}$$

$$\text{so } \vec{AB} = \vec{PS} - (\vec{a} - \vec{b}) \quad \dots \quad (1)$$

$$\text{and from i)} \quad \vec{AB} = \vec{QR} - (\vec{b} - \vec{a}) \quad \dots \quad (2)$$

$$(1) + (2)$$

$$2\vec{AB} = \vec{PS} + \vec{QR} - (\vec{a} - \vec{b}) - (\vec{b} - \vec{a}) \quad (9)$$

$$2\vec{AB} = \vec{PS} + \vec{QR} \quad -a + b - b + a$$

$$2\vec{AB} = \vec{PS} + \vec{QR}$$

$$\vec{AB} = \frac{1}{2}(\vec{PS} + \vec{QR})$$

12/ i) $R \sin(x+\alpha) = \sqrt{3} \sin x + 1 \cos x$

$$R \sin x \cos \alpha + R \cos x \sin \alpha = \sqrt{3} \sin x + 1 \cos x$$

$$(R \cos \alpha) \sin x + (R \sin \alpha) \cos x = \sqrt{3} \sin x + 1 \cos x$$

$$\therefore R \sin \alpha = 1 \quad \text{--- (1)}$$

$$R \cos \alpha = \sqrt{3} \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2} \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \quad \text{since } 0 < \alpha < \frac{\pi}{2}$$

$$\textcircled{1}^2 + \textcircled{2}^2 \quad R^2(\sin^2 \alpha + \cos^2 \alpha) = 1 + 3$$

$$R^2 = 4$$

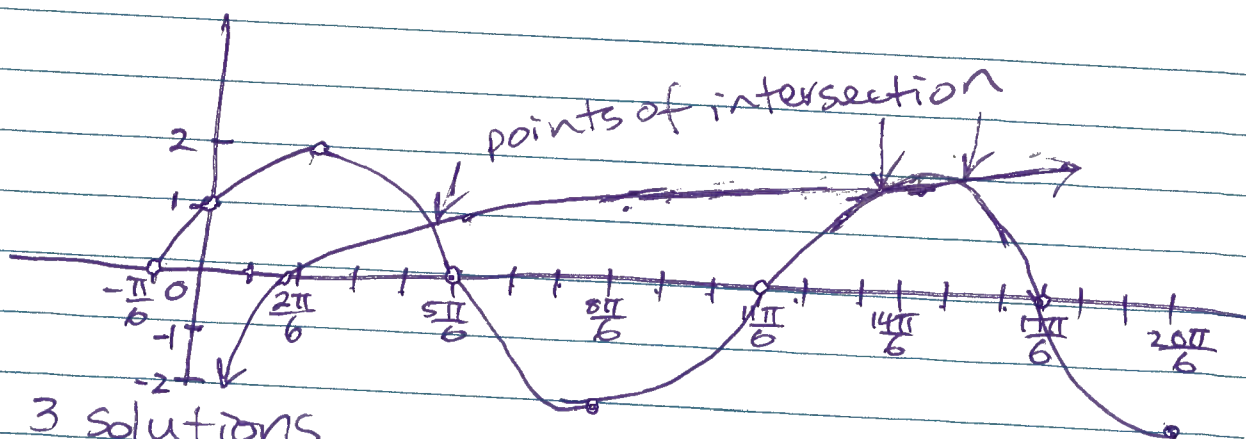
$$R = \pm 2 \quad \therefore R = 2 \quad \text{since } R > 0$$

so $\sqrt{3} \sin x + \cos x = 2 \sin(x + \frac{\pi}{6})$

ii) Missing numbers are

1 1.99 2.19

iii)



iv) 3 solutions

(12)

$$b) (1+x)^m \Rightarrow {}^m C_r x^r$$

$$(1+x)^n \Rightarrow {}^n C_r x^r$$

so to get x^4 we need on LHS

$${}^m C_0 x^0 {}^n C_4 x^4 + {}^m C_1 x^1 {}^n C_3 x^3 + {}^m C_2 x^2 {}^n C_2 x^2$$

$$+ {}^m C_3 x^3 {}^n C_1 x^1 + {}^m C_4 x^4 {}^n C_0 x^0$$

ie $({}^m C_0 {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4 {}^n C_0) x^4$

On RHS

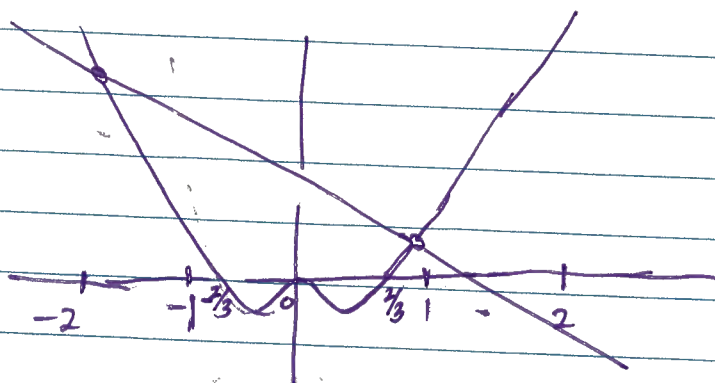
$${}^{m+n} C_4 x^4$$

$$\therefore {}^{m+n} C_4 = {}^m C_0 {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4 {}^n C_0$$

$$= {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4$$

$$= {}^n C_4 + {}^n C_3 {}^m C_1 + {}^n C_2 {}^m C_2 + {}^n C_1 {}^m C_3 + {}^m C_4$$

c) i)



for x-intercepts

$$3x^3 - 2x^2 = 0$$

$$x^2(3x-2) = 0$$

$$x = 0, \frac{2}{3}$$

$\therefore$ by reflection $-\frac{2}{3}$

ii)

$$y = 3(1)^3 - 2(1)^2 = 1 \Rightarrow \text{intersect at } (1, 1)$$

$$y = 3(-2)^3 - 2(-2)^2 = 16 \Rightarrow \text{intersect at } (-2, 16)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{16 - 1}{-2 - 1} (x - 1)$$

$$y - 1 = -5(x - 1)$$

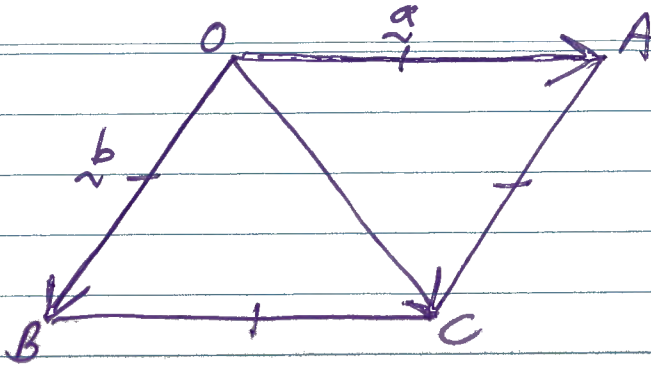
$$y = -5x + 5 + 1$$

$$y = -5x + 6$$

$$\therefore m = -5, b = 6$$

$$d) \quad 6 \times 5 \times 4 = 120 \quad ({}^6P_3)$$

13/



$$\begin{aligned}\vec{OC} &= \vec{OA} + \vec{AC} \\ &= \vec{OA} + \vec{OB} \\ &= \underline{a} + \underline{b}\end{aligned}$$

$$\begin{aligned}\vec{OC} \cdot \vec{OA} &= (\underline{a} + \underline{b}) \cdot \underline{a} \\ &= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} \\ &= |\underline{a}|^2 + |\underline{a}||\underline{b}|\cos\angle AOB\end{aligned}$$

but $|\underline{a}| = |\underline{b}| \quad \therefore \vec{OC} \cdot \vec{OA} = |\underline{a}|^2 + |\underline{a}|^2 \cos\angle AOB$

$$\begin{aligned}\vec{OC} \cdot \vec{OB} &= (\underline{a} \cdot \underline{b}) \cdot \underline{b} & \text{①} \\ &= \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b} \\ &= |\underline{a}||\underline{b}|\cos\angle AOB + |\underline{b}|^2\end{aligned}$$

but $|\underline{a}| = |\underline{b}| \quad \therefore \vec{OC} \cdot \vec{OB} = |\underline{a}|^2 + |\underline{a}|^2 \cos\angle AOB$

②

From ① + ②

$$\begin{aligned}\vec{OC} \cdot \vec{OA} &= \vec{OC} \cdot \vec{OB} \\ |\vec{OC}||\underline{a}|\cos\angle AOC &= |\vec{OC}||\underline{b}|\cos\angle BOC\end{aligned}$$

but $|\underline{a}| = |\underline{b}|$

⑬

$$|\vec{OC}| |\vec{a}| \cos LAOC = |\vec{OC}| |\vec{a}| \cos LBOC$$

$$\therefore \cos LAOC = \cos LBOC$$

Now both LAOC and LBOC are in the range from 0 to 180° .

$$\therefore LAOC = LBOC$$

$$\begin{aligned} \text{b) } V &= \pi \int_0^h \left(\frac{3}{\sqrt{9+x^2}} \right)^2 dx \\ &= \pi \int_0^h \frac{9}{9+x^2} dx \\ &= 9\pi \int_0^h \frac{1}{9+x^2} dx \\ &= 9\pi \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^h \\ &= 3\pi \left[\tan^{-1} \frac{h}{3} - \tan^{-1} \frac{0}{3} \right] \\ &= 3\pi \tan^{-1} \left(\frac{h}{3} \right) \quad \text{cubic units} \end{aligned}$$

$$\text{c) } \frac{dh}{dt} = 3 \text{ cm/s} \quad \text{find } \frac{dV}{dt} \text{ when } h=6$$

$$\begin{aligned} V &= 3\pi \tan^{-1} \left(\frac{h}{3} \right) \\ \frac{dV}{dh} &= 3\pi \cdot \frac{\frac{1}{3}}{1 + \left(\frac{h}{3} \right)^2} \end{aligned}$$

$$\frac{dv}{dh} = 3\pi \frac{\frac{3}{2}}{\frac{9+h^2}{9}}$$

$$= \frac{9\pi}{9+h^2}$$

$$\text{When } h=6 \quad \frac{dv}{dh} = \frac{9\pi}{9+6^2} = \frac{9\pi}{45} = \frac{\pi}{5}$$

$$\begin{aligned} \text{Now } \frac{dv}{dt} &= \frac{dv}{dh} \times \frac{dh}{dt} \\ &= \frac{\pi}{5} \times 3 \\ &= 0.6\pi \text{ cm}^3/\text{s} \end{aligned}$$

$$\begin{aligned} \text{c) } S &= \frac{2000}{1+199e^{-0.4t}} \\ &= 2000(1+199e^{-0.4t})^{-1} \end{aligned}$$

$$\frac{ds}{dt} = 2000 \times (-1) \times 199 \times (-0.4)e^{-0.4t} \times (1+199e^{-0.4t})^{-2}$$

$$= \frac{(-2000 \times 199 \times -0.4)e^{-0.4t}}{(1+199e^{-0.4t})^2}$$

$$= \frac{2000 \times (0.4 \times 199 e^{-0.4t})}{(1+199e^{-0.4t}) \times (1+199e^{-0.4t})}$$

$$= S \times \frac{\frac{2}{5} \times 199 e^{-0.4t}}{(1+199e^{-0.4t})}$$

$$= \frac{S}{5} \left[\frac{2 \times 199 e^{-0.4t}}{1+199e^{-0.4t}} \right]$$

$$= \frac{S}{5} \left[\frac{2 \times 1000}{1000} \times \frac{199e^{-0.4t}}{1+199e^{-0.4t}} \right]$$

$$= \frac{S}{5} \left[\frac{2000}{1+199e^{-0.4t}} \times \frac{199e^{-0.4t}}{1000} \right]$$

$$= \frac{S}{5} \left[S \times \frac{199e^{-0.4t}}{1000} \right]$$

$$= \frac{S}{5} \left[\frac{S}{1000} \times 199e^{-0.4t} \right]$$

Now, from $S = \frac{2000}{1+199e^{-0.4t}}$ we get

$$S(1+199e^{-0.4t}) = 2000$$

$$1+199e^{-0.4t} = \frac{2000}{S}$$

$$199e^{-0.4t} = \frac{2000}{S} - 1$$

$$= \frac{S}{5} \left[\frac{S}{1000} \times \left(\frac{2000}{S} - 1 \right) \right]$$

$$= \frac{S}{5} \left[2 - \frac{S}{1000} \right]$$

ii) When $t=0$, $S = \frac{2000}{1+199} = 10$

as $t \rightarrow -\infty$ $S \rightarrow 0$

as $t \rightarrow \infty$ $S \rightarrow 2000 \therefore B$

($t=0, 2000$ are given by horizontal slope line segments since they are constant solutions.)

iii) When does $\frac{ds}{dt}$ reach a maximum?

$$\begin{aligned}\frac{ds}{dt} &= \frac{s}{5} \left(2 - \frac{s}{1000} \right) \\ &= \frac{2s}{5} - \frac{s^2}{5000}\end{aligned}$$

$$= -\frac{s^2}{5000} + \frac{2s}{5}$$

$$= \left(-\frac{1}{5000} \right) s^2 + \frac{2}{5} s$$

This is a parabola (quadratic)
Since $a < 0$ it is concave down.

$$\text{Max when } s = -\frac{b}{2a} = \frac{-\frac{2}{5}}{2\left(-\frac{1}{5000}\right)}$$

$$= -\frac{2}{5} \div -\frac{2}{5000}$$

$$= -\frac{2}{5} \times \frac{5000}{-2}$$

$$= 1000$$

$$\text{When } s = 1000, \quad 1000 = \frac{2000}{1 + 199e^{-0.4t}}$$

$$1 + 199e^{-0.4t} = 2$$

$$199e^{-0.4t} = 1$$

$$e^{-0.4t} = \frac{1}{199}$$

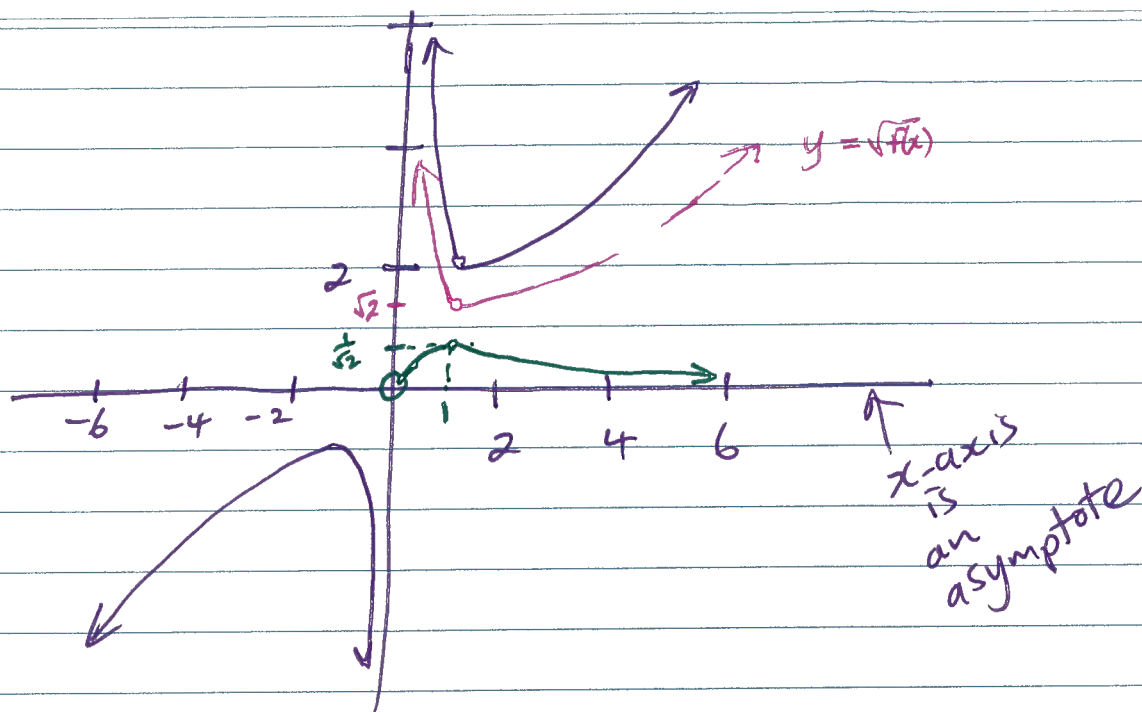
$$e^{0.4t} = 199$$

$$0.4t = \ln 199$$

$$t = \frac{\ln 199}{0.4} = 13.233 \quad (17)$$

About
13 days

14/



$$f(x) = \frac{x^2 + 1}{x}$$

$$f'(x) = \frac{x \cdot 2x - (x^2 + 1) \cdot 1}{x^2} = \frac{2x^2 - x^2 - 1}{x^2} = \frac{x^2 - 1}{x^2}$$

$$\text{for T.P. } \frac{x^2 - 1}{x^2} = 0$$

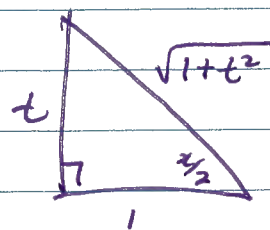
$$\therefore x = 1 \text{ or } -1$$

Can't $\sqrt{\quad}$ negative so we won't consider LHS of y-axis part of original curve.

$\therefore x = 1$ gives min T.P. on original & max T.P. on final curve.

b) i) let $t = \tan \frac{x}{2}$

$$\begin{aligned} \therefore \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} \left(\frac{\sqrt{1+t^2}}{1} \right)^2 \\ &= \frac{1+t^2}{2} \end{aligned}$$



$$\begin{aligned} \frac{dx}{dt} &= \frac{2}{1+t^2} \\ dx &= \frac{2dt}{1+t^2} \end{aligned}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \int \frac{1}{5+3\cos x} dx &= \int \frac{1}{5+3 \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{1}{\frac{5+5t^2+3-3t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{1}{8+2t^2} \cdot 2dt \\ &= \int \frac{1}{4+t^2} \cdot dt \\ &= \frac{1}{2} \tan^{-1} \frac{t}{2} + C \\ &= \frac{1}{2} \tan^{-1} \left(\frac{\tan \frac{x}{2}}{2} \right) + C \\ &= \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \tan \left(\frac{x}{2} \right) \right) + C \end{aligned}$$

$$\begin{aligned}
 \text{ii) Area} &= \int_0^{\pi} \frac{1}{5+3\cos x} \cdot dx \\
 &= \left[\frac{1}{2} \tan^{-1} \left(\frac{1}{2} \tan \frac{x}{2} \right) \right]_0^{\pi} \\
 &= \frac{1}{2} \left[\tan^{-1} \left(\frac{1}{2} \tan \frac{\pi}{2} \right) - \tan^{-1} \left(\frac{1}{2} \tan \frac{0}{2} \right) \right] \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - 0 \right] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

But $\tan \frac{\pi}{2}$ is actually undefined.
 So:

$$\begin{aligned}
 A &= \int_0^{\infty} \frac{1}{4+t^2} dt \quad \leftarrow \text{when } x=\pi, t=\tan \frac{x}{2} \\
 &= \left[\frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) \right]_0^{\infty} \\
 &= \frac{1}{2} \tan^{-1}(\infty) - \frac{1}{2} \tan^{-1} 0 \\
 &= \frac{1}{2} \left(\frac{\pi}{2} \right) - 0 \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$c) i) \quad \underline{a}(t) = 0 \underline{i} - 10 \underline{j}$$

$$\underline{v}(t) = \int \underline{a}(t) \cdot dt$$

$$\underline{v}(t) = -10t \underline{j} + c$$

$$\text{when } t=0, \quad \underline{v}(t) = 90 \cos 30^\circ \underline{i} + 90 \sin 30^\circ \underline{j}$$

$$= 90 \cdot \frac{\sqrt{3}}{2} \underline{i} + 90 \cdot \frac{1}{2} \underline{j}$$

$$= 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\text{so } \underline{v}(0) = -10t \underline{j} + c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$-10(0) \underline{j} + c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\underline{v}(t) = -10t \underline{j} + 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\underline{v}(t) = 45\sqrt{3} \underline{i} + (45-10t) \underline{j}$$

$$\underline{s}(t) = \int \underline{v}(t) dt$$

$$= \int [45\sqrt{3} \underline{i} + (45-10t) \underline{j}] dt$$

$$= 45\sqrt{3}t \underline{i} + (45t - 5t^2) \underline{j} + c$$

$$\text{when } t=0, \quad \underline{s}(t) = 0 \underline{i} + 0 \underline{j}$$

$$\text{so } \underline{s}(0) = 45\sqrt{3}(0) \underline{i} + (45(0) - 5(0)^2) \underline{j} + c = 0 \underline{i} + 0 \underline{j}$$

$$c = 0$$

$$\therefore \underline{s}(t) = 45\sqrt{3}t \underline{i} + (45t - 5t^2) \underline{j} \quad (21)$$

ii) For the stone $\underline{a}(t) = -10 \underline{j}$

$$\underline{x}(t) = V \underline{i} - 10t \underline{j}$$

$$\underline{s}(t) = Vt \underline{i} - 5t^2 \underline{j} + 20 \underline{j}$$

$$\underline{s}(t) = Vt \underline{i} + (20 - 5t^2) \underline{j}$$

Objects collide when position vectors are equal

BUT the stone wasn't fired until 5 seconds after the golf ball was hit.
 $\therefore$ for the stone, relative to the golf-ball

$$\underline{s}(t) = V(t-5) \underline{i} + (20 - 5(t-5)^2) \underline{j}$$

Now we can equate the position vectors of the golf-ball and the stone

$$V(t-5) \underline{i} + (20 - 5(t-5)^2) \underline{j} = (45\sqrt{3}t) \underline{i} + (45t - 5t^2) \underline{j}$$

$$\therefore V(t-5) = 45\sqrt{3}t \text{ and } 20 - 5(t-5)^2 = 45t - 5t^2$$

(equating components)

Using the $\underline{j}$ components:

$$20 - 5(t-5)^2 = 45t - 5t^2$$

$$20 - 5[t^2 - 10t + 25] = 45t - 5t^2$$

$$20 - 5t^2 + 50t - 125 = 45t - 5t^2$$

$$50t - 105 = 45t$$

$$5t = 105$$

$$t = 21 \text{ s}$$

iii) When $t=21$ the $\hat{i}$ components of the position vectors are equal

$$\therefore V(t-5) = 45\sqrt{3} t$$

$$V(21-5) = 45\sqrt{3} (21)$$

$$V = \frac{(45\sqrt{3}(21))}{16}$$

$$V = \frac{945\sqrt{3}}{16} \text{ ms}^{-1}$$

So for the stone

$$V(t) = V\hat{i} - 10t\hat{j}$$

$$V(t) = \frac{945\sqrt{3}}{16} \hat{i} - 10t\hat{j}$$

$$V(16) = \frac{945\sqrt{3}}{16} \hat{i} - 10(16)\hat{j}$$

$$V(16) = \frac{945\sqrt{3}}{16} \hat{i} - 160\hat{j}$$

$$\text{The speed is } |V(21)| = \sqrt{\left(\frac{945\sqrt{3}}{16}\right)^2 + (160)^2}$$

$$= 189.9$$

190 ms^{-1} to 3 significant figures.